

Marissa and Mykuria are planning a trip to Orlando to celebrate graduation. They want to buy a *three*-day ticket to their favorite theme park, which costs \$299 per person but they also need to find a place to stay during their trip. They found a beautiful vacation rental in Orlando that has room for a number of guests and costs a total of \$871.50. If Marissa and Mykuria are able to fill the room up to its maximum capacity with their friends, the price that each person would have to pay for the entire trip would be \$361.25.

1. Define a variable to represent the unknown.
 x represents number of people going on this trip.
2. Write an expression for the 3-day ticket rate per person plus the total rental cost.
 $299x + 871.50$
3. Write another expression for the total amount paid per person for the trip, assuming Marissa and Mykuria are able to get enough friends to come on the trip, so that the vacation rental house will be at maximum capacity.
 $361.25x$
4. If the rental house is at maximum capacity, the *two* expressions will be equivalent to each other. Set the *two* expressions equal to each other to create an equation, then solve your equation.
 $299x + 871.50 = 361.25x$
 $871.50 = 62.25x$
 $14 = x$
5. Interpret your solution in the context of the problem.
14 total people need to come on vacation in order for the price paid per person to be \$361.25.
6. Explain why the *two* expressions would NOT be equivalent if only 6 people attended the trip.
In the expression $361.25x$, the coefficient 361.25 represents the price each person would have to pay only if there is enough people to fill up the room. Since the room can fit over 6, the price per person will be greater than if the room were full.

Javier is in a bowling league. On Friday nights he bowls at Pinz Bowling Alley where it costs \$11.47 per game and another \$9.55 to rent bowling shoes. On Saturday nights he bowls at Vista Bowl where games cost \$12.50 each, shoe rentals cost \$3.95, and drinks cost \$1.48. This past Saturday at Vista Bowl, Javier bowled the same number of games he bowled the night before at Pinz. He also spent the same amount of money at Vista Bowl that he spent at Pinz, even though he also purchased one drink at Vista Bowl. How many games did Javier bowl at each bowling alley?

7. Define a variable to represent the unknown.

Answers may vary slightly.

x represents number of games Javier bowled

8. Write and solve an equation that represents this situation.

$$11.47x + 9.55 = 12.5x + 3.95 + 1.48$$

$$1.03x = 4.12$$

$$x = 4$$

9. Interpret your solution.

Javier bowled 4 games at each bowling alley.

10. Why is 6.5 not a viable solution for this context?

Games bowled needs to be an integer since he can't pay for 0.5 games.